SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1506

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I _M.Mary Sharlin (192511330)___(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: _M.Mary Sharlin_

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

ANSWERS:

1.

A company experiences heavy traffic during seasonal sales. Cloud computing uses auto-scaling and load balancing to handle this efficiently.

Auto-Scaling

➔ Auto-scaling automatically increases or decreases the number of servers based on user traffic.
➔ During peak sales, new virtual machines are added automatically.
➔ When traffic decreases, extra servers are removed to reduce cost.

Load Balancing

➔ Load balancing distributes incoming user requests equally among multiple servers.
➔ It prevents a single server from becoming overloaded.
➔ If one server fails, requests are redirected to healthy servers.

Reducing Downtime

➔ Traffic is evenly distributed across servers.
➔ Automatic scaling prevents server crashes.
➔ High availability ensures websites remain accessible.

Improving User Experience

➔ Faster page loading.
➔ Quick response during checkout.
➔ Reduced waiting time.
➔ Smooth shopping experience.

Role of Monitoring Tools

Monitoring tools continuously observe:

➔ CPU usage
➔ Memory utilization
➔ Network traffic
➔ Response time

If usage exceeds a threshold, auto-scaling is triggered automatically.

Real-World Example

During Amazon's Great Indian Festival or Flipkart Big Billion Days, millions of customers visit simultaneously. Cloud auto-scaling adds extra servers while load balancers distribute requests, ensuring uninterrupted shopping.

2.

Organizations store large backup data securely using cloud storage.

Data Durability

Cloud providers store multiple copies of data across different storage devices.

Benefits:

- Prevents data loss.
- Maintains data integrity.
- Ensures long-term availability.

Data Redundancy

Data is replicated across multiple regions or data centers.

Advantages:

- Protection against hardware failure.
- Recovery during disasters.
- Continuous service availability.

Encryption

Encryption converts data into unreadable form.

Two types:

- Data at Rest Encryption
- Data in Transit Encryption

Only authorized users with proper keys can access data.

Access Control

Cloud providers implement:

- User authentication
- Multi-factor authentication (MFA)
- Role-Based Access Control (RBAC)

These prevent unauthorized access.

Cost Advantages

Compared to traditional storage:

- No need to buy storage hardware.
- No maintenance costs.
- Pay only for storage used.
- Easy scalability.

Disaster Recovery Example

If a company's local server is damaged due to fire or flood, cloud backups stored in another region can quickly restore all business data, ensuring business continuity.

3.

A mobile application deployed worldwide requires fast access for users.

Content Delivery Network (CDN)

A CDN is a network of distributed servers that stores cached copies of application content closer to users.

Benefits

- Faster content delivery.
- Reduced network congestion.
- Lower latency.
- Higher availability.

Edge Servers

Edge servers are located near users.

Functions:

- Deliver cached content.
- Reduce travel distance for data.
- Improve response time.

Cloud Global Availability

Cloud providers maintain multiple data centers across different countries.

Advantages:

- Global accessibility.
- Automatic failover.
- High availability.
- Better disaster recovery.

Reduced Latency

Since users connect to the nearest edge server, delay is minimized.

Real-Time Example

Netflix and YouTube use CDNs to stream videos smoothly worldwide. Similarly, online games like PUBG and Free Fire use edge servers to reduce lag and provide real-time gameplay.

4.

Financial institutions require strong security and regulatory compliance.

Regulatory Compliance

Cloud providers support standards such as:

- ISO 27001
- PCI DSS
- GDPR
- HIPAA

These help organizations meet legal and industry requirements.

Auditing

Cloud platforms maintain audit logs that record:

- User login
- File access
- Configuration changes
- Security events

Auditing helps detect suspicious activities.

Logging

Logging records every system activity.

Benefits:

- Security analysis.
- Troubleshooting.
- Compliance reporting.

Monitoring

Monitoring tools continuously check:

- System health
- Network traffic
- Unauthorized access
- Resource utilization

Alerts are generated when unusual activity is detected.

Encryption

Sensitive financial information is protected using:

- Encryption at Rest
- Encryption in Transit

Even if data is intercepted, it cannot be read.

Data Sovereignty

Financial organizations can store data only within specific countries or regions to comply with government regulations.

Cloud providers allow customers to choose storage locations.

Real-World Example

A bank processes online transactions using encrypted cloud storage, audit logs, continuous monitoring, and regional data centers to ensure secure financial transactions and regulatory compliance.

5.

Development teams use containers for efficient application deployment.

Containers vs Virtual Machines

Containers	Virtual Machines (VMs)
Share the host operating system	Each VM has its own operating system
Lightweight	Heavyweight

Fast startup	Slow startup
Lower resource usage	Higher resource usage
Higher performance	More overhead

Benefits of Containers

- Fast deployment.
- Efficient resource utilization.
- Easy application updates.
- Consistent environment.

Kubernetes Orchestration

Kubernetes automates:

- Container deployment.
- Scaling.
- Load balancing.
- Self-healing.
- Resource management.

If a container fails, Kubernetes automatically creates a new one.

Portability

Containers run consistently across:

- Private cloud
- Public cloud
- Hybrid cloud
- Local systems

This avoids compatibility issues.

Microservices Example

An e-commerce application can be divided into:

- User Service
- Product Service
- Payment Service
- Order Service
- Notification Service

Each service runs in its own container. Kubernetes manages deployment, scaling, and recovery automatically, making the application reliable and easy to maintain.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity